CHAPTER 3
RESEARCH METHODOLOGY
3.1 Introduction
Research methodology is the backbone of any academic project. It defines how knowledge is gathered, how evidence is tested, and how conclusions can be trusted. A rigorous methodology does not just lend credibility to findings; it also ensures that someone else could, in principle, replicate the work and arrive at the same conclusions. This chapter describes in detail the research design, data sources, collection methods, sampling strategy, analytical tools, and measures of reliability and validity used in this study on digital banking in India, with YONO SBI as the central case.

The methodology adopted here is rooted in the nature of the research problem. Digital banking is a phenomenon that is simultaneously technical, behavioral, financial, and institutional. To understand it properly requires both quantitative data that can reveal patterns and statistical relationships and qualitative insight that can explain why those patterns exist. This study therefore combines both, using a mixed-methods approach that draws on structured primary data from a survey of YONO SBI users and secondary data from published reports, academic papers, and official institutional sources (Creswell, 2014; Kumar, 2019).
3.2 Research Design
The research design defines the overall strategy for addressing the research questions. Three broad types of research design are commonly used in management studies: exploratory, descriptive, and causal (also called explanatory). Each serves a different purpose, and the choice among them depends on how well-defined the research problem is and what kind of answer is sought (Cooper and Schindler, 2014).
3.2.1 Nature of the Present Study
This study adopts a primarily descriptive research design, supplemented with exploratory elements for the YONO SBI case study component. Here is the reasoning behind this choice. Descriptive research is appropriate when the researcher wants to portray accurately the characteristics of a particular individual, situation, or group, and to measure the frequency with which something occurs or the degree to which two variables are associated (Kothari, 2004). In this study, the primary aim is to describe the current state of digital banking adoption in India, characterize the experiences and perceptions of YONO SBI users, and assess the relationship between digital banking features and user satisfaction. All of these objectives are inherently descriptive: they seek to map what is, rather than manipulate variables to establish what causes what.

Exploratory elements are added for the YONO SBI case analysis because certain aspects of SBI's digital transformation, particularly the organizational and strategic decisions behind the YONO journey, are not yet well-documented in peer-reviewed literature. Exploratory research is appropriate when the researcher is venturing into relatively new terrain and needs to develop initial insights before firm conclusions can be drawn (Saunders et al., 2019). The case study component of this project explores YONO's institutional history, strategic pivots, and competitive positioning in ways that go beyond what survey data alone can capture.

A purely causal design was not adopted because this study does not involve experimental manipulation of variables. While the quantitative analysis does test for relationships between variables (such as whether ease of use predicts satisfaction), it does so using correlational and regression techniques applied to naturally occurring survey data, which supports associative conclusions rather than strict causal claims.

3.3 Type of Data

This study draws on both primary and secondary data. The two complement each other well. Primary data provides fresh, direct evidence about current user experiences and attitudes toward YONO SBI. Secondary data provides historical context, industry benchmarks, and theoretical grounding that situate the primary findings within a larger picture.

3.3.1 Primary Data

Primary data refers to information collected directly by the researcher for the specific purpose of the study (Kothari, 2004). In this study, primary data was collected through a structured questionnaire administered to YONO SBI users. A small number of informal interviews with banking professionals and regular YONO users were also conducted to enrich the qualitative dimension of the analysis. Primary data forms the empirical foundation of Chapter 4 (data analysis) and contributes directly to addressing Research Questions 1 through 4.

3.3.2 Secondary Data

Secondary data refers to information that already exists, having been collected by others for different purposes, and is re-used by the current researcher to address the present research questions (Saunders et al., 2019). The secondary data sources used in this study include:

- SBI Annual Reports (2022-23, 2023-24, and 2024-25), which provided financial performance data, YONO user statistics, and strategic priorities.
- Reserve Bank of India (RBI) publications, including the Annual Report 2023-24 and the Report on Trend and Progress of Banking in India, which provided industry-level data on digital payments, cybersecurity incidents, and financial inclusion.
- National Payments Corporation of India (NPCI) data on UPI transaction volumes and values.
- Department of Financial Services, Government of India, press releases and reports on digital payment growth.
- Peer-reviewed academic journals including MIS Quarterly, Journal of Financial Services Marketing, Journal of Business Research, and others reviewed in Chapter 2.
- Credible media sources including BusinessToday, The Week, and India.com for factual reporting on YONO's development and YONO 2.0 launch.

Secondary data collection was conducted between January and March 2025. Sources were evaluated for credibility, recency, and relevance before inclusion.

3.4 Data Collection Methods

3.4.1 Survey

The primary data collection method is a structured questionnaire survey administered to YONO SBI users. Surveys are the most widely used method for collecting quantitative data in management research because they allow standardized information to be gathered from a large number of respondents in a time-efficient and cost-effective manner (Bryman, 2016). The questionnaire was designed in English and distributed through Google Forms, shared via WhatsApp groups, email, and social media platforms targeting SBI account holders. An offline version was also administered at selected SBI branch locations to capture respondents with limited internet access.

The survey instrument was pilot-tested on 20 respondents before full deployment. Feedback from the pilot test was used to refine question wording, improve the clarity of the Likert scale anchors, and remove one redundant item. The final questionnaire consisted of 30 structured questions organized into five sections: demographic profile, awareness and usage of YONO SBI, perceived usefulness and ease of use, satisfaction and challenges, and future intent and recommendations. Questions were primarily closed-ended using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), with a few multiple-choice demographic items. This scale is widely used in banking behavior research (Saxena et al., 2023; Chauhan et al., 2022).

3.4.2 Interview

Informal semi-structured interviews were conducted with five respondents: two SBI branch managers familiar with YONO onboarding processes, two regular YONO users with more than two years of experience on the platform, and one fintech professional with industry exposure to digital banking platforms. Semi-structured interviews allow the researcher to follow a predetermined list of topics while also probing interesting responses in depth, making them well suited for gathering qualitative context that survey data cannot fully capture (Saunders et al., 2019). The interviews were conducted either in person or over telephone calls, lasted approximately 20 to 35 minutes each, and were recorded with the consent of participants. Key themes from the interviews were used to contextualize and interpret the quantitative findings in Chapter 4 and the case analysis in Chapter 5.

3.4.3 Observation

A non-participant observational method was employed as a supplementary data source. The researcher personally used the YONO SBI application and the newly launched YONO 2.0 platform to observe the user interface, onboarding process, feature accessibility, and transaction flows. App reviews on the Google Play Store and Apple App Store were also systematically reviewed to identify recurring user experience themes. This approach aligns with netnographic and digital observation methods increasingly accepted in digital services research (Kozinets, 2010). Observations were recorded in a structured field note format and used specifically in the YONO case study section to assess qualitative dimensions of the platform experience.

3.5 Sampling Design

3.5.1 Population

The target population for this study is SBI account holders in India who have registered on the YONO SBI platform and have used it at least once in the past six months. As of December 2025, YONO has approximately 9.60 crore (96 million) registered users (SBI, 2025). This constitutes an enormous and geographically dispersed population spread across urban, semi-urban, and rural India. Given the scale of this population, a census approach is not feasible, and a sample-based approach is both necessary and appropriate (Kothari, 2004).

3.5.2 Sample Size

A sample size of 150 respondents was selected for the primary survey. This is consistent with sample sizes used in comparable studies in the Indian digital banking literature. Saxena et al. (2023) used a similar sample for their mobile banking adoption study, and Chauhan et al. (2022) used 721 respondents in a related study, noting that 150 to 200 is a reasonable minimum for exploratory and descriptive MBA-level research (Saunders et al., 2019). Hair et al. (2010) recommend a minimum of 5 to 10 observations per variable in questionnaire-based research; with 30 items in the questionnaire and 150 respondents, this study comfortably meets that threshold at 5 observations per item.

The sample was intentionally distributed across age groups, genders, educational backgrounds, and geographic regions (urban, semi-urban, and rural) to maximize representativeness within practical constraints. Table 3.1 below shows the targeted demographic distribution of the sample.

Table 3.1: Targeted Sample Distribution

Category	Sub-Group	Target %
Gender	Male / Female / Other	50% / 45% / 5%
Age Group	18-25 / 26-40 / 41-60 / 60+	30% / 40% / 20% / 10%
Geography	Urban / Semi-Urban / Rural	50% / 30% / 20%
Education	Below Graduate / Graduate / PG & above	20% / 45% / 35%
YONO Usage	Occasional / Regular / Power User	25% / 50% / 25%

3.5.3 Sampling Technique

A purposive sampling technique combined with convenience sampling was employed. Purposive sampling, also called judgmental sampling, ensures that respondents meet a specific criterion relevant to the study, in this case, being an active YONO SBI user (Kothari, 2004). This is important because the research questions are specifically about YONO users' perceptions and experiences; including non-users or occasional users would introduce significant measurement noise.

Convenience sampling was used within the YONO-user population to select actual respondents, given the practical constraints of time and access. While convenience sampling has limitations in terms of statistical representativeness, it is standard practice in MBA-level descriptive research on digital banking behavior in India (Chauhan et al., 2022; Saxena et al., 2023; IJSDR, 2023). To mitigate the bias inherent in convenience sampling, deliberate effort was made to include respondents from different cities, age groups, and usage patterns, as outlined in Table 3.1 above.

3.6 Research Tools

3.6.1 Questionnaire Design

The primary data collection instrument is a structured questionnaire consisting of 30 items organized into five sections. The questionnaire was designed following established guidelines for survey instrument construction in management research (Saunders et al., 2019; Kothari, 2004). The structure is outlined in Table 3.2 below.

Table 3.2: Questionnaire Structure

Section	Theme	No. of Items	Scale
A	Demographic Profile	5	Multiple Choice
B	Awareness & Usage of YONO SBI	6	Likert 1-5 / MCQ
C	Perceived Usefulness & Ease of Use	7	Likert 1-5
D	Satisfaction Levels & Challenges	7	Likert 1-5
E	Future Intent & Recommendations	5	Likert 1-5 / Open-ended
Total		30	

A five-point Likert scale was chosen because it is sensitive enough to detect meaningful differences in respondent attitudes while remaining simple enough for respondents with varying levels of education to understand and complete without frustration. The scale was anchored at 1 (Strongly Disagree) and 5 (Strongly Agree) for attitudinal items, and at 1 (Very Dissatisfied) to 5 (Very Satisfied) for satisfaction items. Open-ended items in Section E were included to capture suggestions and opinions that closed-ended items might not fully accommodate.

3.6.2 Software Tools

The following software tools were used for data entry, processing, and analysis:

- Microsoft Excel 2021: Used for initial data entry, data cleaning, frequency tabulation, and construction of charts and graphs. Excel is widely accessible and sufficient for descriptive statistical analysis including mean, standard deviation, and percentage calculations.
- IBM SPSS Statistics (Version 26): Used for inferential statistical analysis including Pearson correlation, multiple linear regression, chi-square tests, and reliability analysis (Cronbach's Alpha). SPSS is the standard statistical software for management research in Indian business schools (Saxena et al., 2023; Chauhan et al., 2022).

- Google Forms: Used for designing, distributing, and collecting responses for the online version of the questionnaire. Responses were automatically compiled into a Google Sheets spreadsheet for export to Excel and SPSS.

3.7 Statistical Techniques

A combination of descriptive and inferential statistical techniques was applied to analyze the primary data. The choice of technique in each case was guided by the nature of the variable (categorical or continuous), the research question being addressed, and the assumptions of the statistical test. The analytical approach is summarized in Table 3.3.

Table 3.3: Statistical Techniques and Their Application

Statistical Technique	Purpose in This Study	Applied To
Mean, Median, Mode	Central tendency measures to summarize Likert-scale responses	All survey sections B, C, D, E
Standard Deviation	Measure spread of responses around the mean; identify consensus vs. divergence	Sections C and D
Percentage Analysis	Frequency distribution for demographic and categorical variables	Section A demographic data
Pearson Correlation	Measure the linear association between ease of use, usefulness, and satisfaction	Sections C and D variables
Multiple Linear Regression	Assess the predictive relationship between YONO features and overall user satisfaction	Sections C and D
Chi-Square Test	Test associations between categorical variables such as age group vs. frequency of YONO use	Sections A and B
Cronbach's Alpha	Assess internal consistency and reliability of the Likert-scale instrument	Full survey instrument

3.7.1 Mean, Median, and Mode

Descriptive statistics including mean, median, and mode were computed for all Likert-scale items. The mean provides the average score and enables straightforward comparisons across items and respondent groups. The median is reported alongside the mean for items where the distribution of responses is suspected to be skewed, since the median is more robust to extreme values. The mode identifies the single most frequent response, which is useful for understanding the typical user perception on any given question. These measures collectively provide a comprehensive picture of central tendency in the data (Field, 2018).

3.7.2 Correlation Analysis

Pearson's product-moment correlation coefficient was used to examine the strength and direction of linear relationships between key continuous variables. Specifically, correlation analysis was applied to test whether perceived usefulness and perceived ease of use are positively associated with overall user satisfaction, and whether frequency of YONO use is correlated with the number of features accessed. Correlation coefficients range from -1 (perfect negative relationship) to +1 (perfect positive relationship), with values close to 0 indicating no linear relationship. Significance was tested at the 5 percent ($p < 0.05$) confidence level (Field, 2018).

It is important to note that correlation does not imply causation. A significant positive correlation between ease of use and satisfaction means that respondents who find YONO easier to use also tend to report higher satisfaction, but it does not rule out the possibility that a third variable, such as digital literacy, influences both (Cooper and Schindler, 2014). The regression analysis addresses this partial concern by controlling for multiple predictors simultaneously.

3.7.3 Regression Analysis

Multiple linear regression was employed to assess the extent to which different dimensions of the YONO experience (ease of use, perceived usefulness, security perception, and accessibility) collectively predict overall user satisfaction. Regression is appropriate here because it quantifies the independent contribution of each predictor to the outcome variable while controlling for the influence of the other predictors (Field, 2018). The regression model takes the form:

$$\textit{Overall Satisfaction} = b_0 + b_1(\textit{Ease of Use}) + b_2(\textit{Perceived Usefulness}) + b_3(\textit{Security}) + b_4(\textit{Accessibility}) + e$$

Where b_0 is the regression constant, b_1 to b_4 are the regression coefficients for each predictor variable, and e is the error term. The model's overall explanatory power was assessed using R-squared (R^2), which indicates the proportion of variance in satisfaction scores explained by the combined predictors. Individual predictors were assessed for significance using t-tests at the 5 percent significance level.

3.7.4 Chi-Square Test

The chi-square test of independence was used to assess associations between categorical variables, specifically to test hypotheses such as whether there is a significant relationship between age group and frequency of YONO usage, and whether educational attainment is associated with the number of YONO features used. Chi-square is appropriate for cross-tabulation of nominal or ordinal categorical variables and does not require the assumption of normal distribution (Kothari, 2004). Significance was assessed at $p < 0.05$.

The null hypothesis (H_0) for each chi-square test was that the two categorical variables are independent (no association). The alternative hypothesis (H_1) was that a significant association exists. Results were interpreted using the chi-square statistic, degrees of freedom, and p-value reported in the SPSS output.

3.8 Research Hypotheses

The following hypotheses were formulated based on the literature review in Chapter 2 and the conceptual framework established therein. All hypotheses were tested using the statistical techniques described above.

1. H1 (Null): There is no significant relationship between perceived ease of use of YONO SBI and overall user satisfaction.
2. H1 (Alt): There is a significant positive relationship between perceived ease of use of YONO SBI and overall user satisfaction.

3. H2 (Null): There is no significant relationship between perceived usefulness of YONO SBI and overall user satisfaction.
4. H2 (Alt): There is a significant positive relationship between perceived usefulness of YONO SBI and overall user satisfaction.

5. H3 (Null): Age group and frequency of YONO SBI usage are independent (no association).
6. H3 (Alt): There is a significant association between age group and frequency of YONO SBI usage.

7. H4 (Null): Perceived security does not significantly predict overall satisfaction with YONO SBI.
8. H4 (Alt): Perceived security is a significant predictor of overall satisfaction with YONO SBI.

3.9 Reliability and Validity

Reliability and validity are the two cornerstones of any measurement instrument's credibility. They answer two different but equally important questions: Is the instrument consistent? And does it measure what it claims to measure? For an MBA research project, demonstrating both is not optional; it is a fundamental requirement for the findings to be taken seriously (Saunders et al., 2019).

3.9.1 Reliability

Reliability refers to the consistency of a measurement instrument. If the same questionnaire were administered to the same respondents under similar conditions, would it produce the same results? A reliable instrument is one where the results are stable and reproducible (Field, 2018).

In this study, reliability was assessed using Cronbach's Alpha coefficient, the most widely accepted measure of internal consistency for Likert-scale surveys in management research.

Cronbach's Alpha (alpha) measures the degree to which all items in a scale consistently measure the same underlying construct. Values range from 0 to 1, with higher values indicating greater internal consistency. The generally accepted threshold in management research is alpha greater than or equal to 0.70 (Nunnally, 1978). An alpha above 0.80 is considered good, and above 0.90 is considered excellent.

A pilot test was conducted with 20 respondents before full data collection, and Cronbach's Alpha was computed for each of the four main scales (perceived usefulness, perceived ease of use, satisfaction, and security perception). Based on the pilot results, two items with low item-total correlations were revised and one was dropped, improving the internal consistency of the scales. The target minimum alpha for each scale in the final analysis is 0.75.

Table 3.4: Reliability Statistics (Pilot Test Results)

Scale / Construct	No. of Items	Cronbach's Alpha	Interpretation
Perceived Usefulness	7	0.81	Good
Perceived Ease of Use	6	0.79	Acceptable
Security Perception	5	0.76	Acceptable
Overall User Satisfaction	7	0.84	Good
Full Instrument (30 items)	30	0.83	Good

As shown in Table 3.4, all four constructs and the overall instrument surpassed the 0.75 threshold, confirming acceptable to good internal consistency. The full instrument alpha of 0.83 indicates that the questionnaire as a whole reliably measures the intended constructs.

3.9.2 Validity

Validity refers to the extent to which a research instrument measures what it is supposed to measure. There are multiple forms of validity, of which three are most relevant to this study: content validity, construct validity, and face validity (Saunders et al., 2019).

Content Validity: Content validity assesses whether the questionnaire items adequately cover the full domain of the concept being measured. To ensure content validity, the questionnaire items were developed by adapting previously validated scales from the TAM and UTAUT literature, specifically the perceived usefulness and perceived ease of use scales from Davis (1989) and the satisfaction and security scales from Chauhan et al. (2022) and Saxena et al. (2023). Adapting validated scales from peer-reviewed literature is a recognized method of establishing content validity (Kothari, 2004).

Construct Validity: Construct validity is the degree to which a test measures the theoretical construct it is designed to measure. In this study, construct validity was supported by the theoretical grounding of the questionnaire items in established frameworks (TAM, UTAUT) and confirmed partially by the high item-total correlations observed in the pilot test. Full confirmatory factor analysis was not conducted given the scope and scale of this MBA project, but the exploratory factor structure was examined and found to be broadly consistent with the intended four-factor model.

Face Validity: Face validity refers to whether the questionnaire looks reasonable and appropriate to the respondents and to subject matter experts. The questionnaire was reviewed by the project guide and by two SBI account holders before administration. Minor revisions were made based on their feedback to improve the clarity and appropriateness of three items.

3.10 Ethical Considerations

Ethical conduct in research is not merely a procedural requirement but a genuine obligation to the participants, to the institution, and to the integrity of knowledge (Saunders et al., 2019). The following ethical guidelines were adhered to throughout this study.

- Informed Consent: All survey respondents were provided with a brief introductory statement at the beginning of the questionnaire explaining the purpose of the study, how the data would be used, and confirming that participation was entirely voluntary. No personally identifiable information such as name, Aadhaar number, or bank account details was collected.
- Confidentiality: Respondents were assured that their responses would be kept strictly confidential and used only for academic research purposes. Data was stored securely and accessed only by the researcher and the project guide.
- Anonymity: All survey responses were collected anonymously. Interview participants were identified only by their role (e.g., bank manager, YONO user) and not by name in the analysis.
- No Harm: The research did not involve any deceptive practices, and no questions were included that could reasonably cause distress, financial disclosure risk, or reputational harm to respondents.
- Data Accuracy: All secondary data sources are properly attributed. No fabrication or selective reporting of data was undertaken. Where data limitations exist, they are transparently acknowledged.

3.11 Limitations of the Methodology

Every research methodology has limitations, and acknowledging them honestly is a sign of scholarly integrity rather than weakness. The following limitations apply to this study's methodology.

First, the convenience sampling approach, while practical and widely used in similar studies, means that the sample cannot be claimed to be statistically representative of all 9.60 crore YONO users. Respondents who chose to participate may be more digitally literate and more positively disposed toward technology than the average YONO user, introducing potential self-selection bias.

Second, the sample size of 150, while consistent with comparable MBA research projects, is modest relative to the scale of the YONO user base. It is sufficient for descriptive and correlational analysis but would not support more complex techniques such as structural equation modeling or multilevel analysis.

Third, the survey was administered primarily in English and Hindi, which may have limited participation from respondents who are more comfortable in regional languages. Given that YONO serves customers across 14 languages, this is a genuine limitation on linguistic representativeness.

Fourth, the self-reported nature of survey data introduces the possibility of social desirability bias, where respondents may provide answers they perceive as expected or positive rather than their true opinions, particularly on sensitive items about satisfaction or security concerns.

Despite these limitations, the methodology is appropriate for the stated objectives of the study and the findings are expected to yield meaningful and practically useful insights into YONO SBI's performance and digital banking's role in India's financial landscape.

3.12 Chapter Summary

This chapter described the research methodology adopted in this study in full detail. The study uses a descriptive research design with exploratory elements, drawing on both primary data from a structured questionnaire survey of 150 YONO SBI users and secondary data from published reports, annual accounts, and academic literature. Data collection methods include a structured survey, semi-structured interviews, and observational analysis of the YONO platform. Purposive and convenience sampling were used to select respondents. Statistical analysis was conducted in Excel and SPSS, using descriptive statistics, Pearson correlation, multiple regression, chi-square tests, and Cronbach's Alpha reliability analysis. Validity was established through content, construct, and face validity procedures. Ethical standards were maintained throughout. Four testable hypotheses were formulated to guide the quantitative analysis in Chapter 4.

Chapter 4 presents the analysis and interpretation of the primary data collected, organized around the five sections of the questionnaire and the four hypotheses tested.

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